FINAL ANALYSIS REPORT

Formula Definition

$$T_{\text{eff}}(T, U\cancel{, H}) = T - \frac{93}{943}U^2 + \frac{500}{739}U + \cancel{\frac{93H}{943 \cdot 1819 \cdot 708}}$$

$$d_{\text{eff}}(d, H, U) = d + \frac{H^{3/4}}{d^{1/4}} + \frac{986}{873}U \tanh(H)$$

Global Accuracy

Metric	Value
MAPE (Mean % Error)	2.2663%
Max Error Percent	70.2379%
RMSE (Average Unit Error)	3.304398
Bias (Mean Residual)	0.292536

Percentile Breakdown

Coverage Level	Absolute Percentage Error
50% of data	< 1.8471%
90% of data	< 4.6112%
99% of data	< 7.5232%

Regime Analysis

Regime	Count	MAPE	Max Error
Shallow	3341	2.4018%	19.2788%
Intermediate	2793	2.1122%	70.2379%
Deep	3482	2.2598%	36.0974%

Top 20 Worst Predictions

Index	H	T	d	U	L True	L Pred	Diff	Error %
4799	2.84	4.72	14.10	-2.35	38.07	11.33	26.74	70.24
1068	1.04	4.90	49.87	-1.91	12.40	16.87	-4.48	36.10
2476	1.04	4.28	43.77	-1.58	11.73	14.09	-2.37	20.18
6850	1.00	19.32	1.68	-2.48	42.79	34.54	8.25	19.28
2772	1.98	4.74	17.61	-1.78	14.32	16.99	-2.67	18.65
2969	1.08	6.08	17.45	-2.35	20.93	24.79	-3.86	18.46
817	1.40	4.49	47.27	-1.69	12.85	15.18	-2.33	18.17
7141	1.00	18.67	1.51	-2.27	42.52	35.13	7.38	17.37

Index	H	T	d	U	L True	L Pred	Diff	Error %
8555	1.00	15.56	1.38	-2.26	30.55	25.32	5.23	17.12
1618	2.21	5.75	45.28	-2.19	20.21	23.50	-3.28	16.24
9303	1.00	15.45	1.37	-2.22	30.96	26.12	4.84	15.64
1199	1.79	4.72	14.46	-1.81	14.19	16.39	-2.20	15.53
2523	2.36	5.62	37.97	-2.14	19.74	22.67	-2.93	14.83
8909	1.00	18.63	1.64	-2.34	43.19	36.84	6.36	14.72
7832	1.00	17.19	1.76	-2.47	38.53	33.11	5.42	14.07
2530	2.17	5.85	25.02	-2.20	21.54	24.57	-3.03	14.05
7652	1.00	15.15	1.43	-2.26	30.11	25.98	4.14	13.73
8558	1.00	15.83	1.58	-2.25	36.47	31.67	4.80	13.15
9600	1.00	14.41	1.43	-2.12	32.62	28.33	4.29	13.14
9339	1.00	17.44	1.49	-2.29	35.95	31.29	4.66	12.96